Ajith Suresh

Lead MPC Researcher

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"Be weird. Be random. Be who you are. Because you never know who would love the person you hide."

Research Interests

My research primarily centers on cryptography and information security, with a special emphasis on creating efficient protocols for problems in the area of Secure Multi-party Computation (MPC). My work revolves around designing and implementing practical MPC protocols, as well as delving into the realms of privacy-preserving machine learning and federated learning. I have actively engaged in a diverse range of cryptographic areas, including Multi-party Computation, Verifiable Secret Sharing, Oblivious Transfer, Byzantine Agreement, Broadcast, as well as cutting-edge domains like Privacy-Preserving Machine Learning and Federated Learning.

Education

Indian Institute of Science (IISc) — *Bengaluru, India*

Ph. D. in Computer Science — *Sep. 2017 - Jul. 2021*

- Dissertation Area: Secure Multi-party Computation (MPC) & Privacy-Preserving Machine Learning (PPML)
- Advisor: Prof. Arpita Patra
- CGPA: 9 / 10 (First Class with Distinction)

Indian Institute of Science (IISc) — *Bengaluru, India*

M.Tech. (Research) in Computer Science — *Aug. 2014 - Jun. 2017*

- Dissertation Area: Secure Multi-party Computation (MPC)
- Advisor: Prof. Arpita Patra
- CGPA: 6.83 / 8 (First Class with Distinction)

College of Engineering, Trivandrum (CET) — *Trivandrum, India*

B.Tech in Computer Science and Engineering — *Jul. 2010 - Apr. 2014*

- CGPA: 8.81 / 10 (First Class with Distinction)

Professional Experience

Technology Innovation Institute (TII) — *Abu Dhabi, UAE*

Lead MPC Researcher — *Apr. 2025 – Present*

Senior MPC Researcher — *May. 2023 – Mar. 2025*

- Associated with the The Cryptography Research Center (CRC).

Technical University (TU) of Darmstadt — *Darmstadt, Germany*

Post-Doctoral Research in Computer Science — *Oct. 2021 - Mar. 2023*

– Area: Privacy-preserving Services On the Internet (PSOTI)

Research Intern — *Nov. 2019*

– Research work under the joint guidance of Thomas Schneider and Arpita Patra.

- Host: Prof. Thomas Schneider at Cryptography and Privacy Engineering (ENCRYPTO) Research Group

Indian Institute of Science (IISc) — *Bengaluru, India*

Research Associate — *Aug. 2021 - Sep. 2021*

- Host: Prof. Arpita Patra

Amazon Development Centre — *Bengaluru, India*

Software Development Engineer (SDE) Intern — *Jul. 2013 - Aug. 2013*

- Project: "Increase the registered and subscribed user base at Amazon" in the International Expansion (Junglee Traffic) Team.

Awards, Scholarships and Achievements

1. Received Runner-Up Distinguished Paper Award at 2nd IEEE Conference on Secure and Trustworthy Machine Learning (IEEE SaTML'24).
2. Awarded commendation certificate for outstanding PhD thesis by Department of Computer Science and Automation at the Indian Institute of Science (IISc), Bangalore.
3. Nominated for the Schmidt Science Fellows 2022 program for post-doctoral research (one among a group of 350 highly accomplished candidates, nominated from 83 of the world's leading universities and institutes).
4. Represented country India in the "Window to the World" session at the 8th Heidelberg Laureate Forum.
5. Selected as one among 225 young researchers to participate in the 8th Heidelberg Laureate Forum.
6. Recipient of Google PhD Fellowship 2019 (one among 53 researchers around the globe).
7. Organiser of Secure Multi-Party Computation: Theory and Practice Workshop at Indian Institute of Science (IISc) from 19th to 22nd January, 2020.
8. Received travel grant to attend Privacy Preserving Machine Learning Workshop, CCS 2019, London.
9. Received travel grant to attend Workshop: Theory and Practice of Secure Multiparty Computation 2016, Aarhus University, Denmark.
10. Secured All India Rank of 807 with a score of 688 in GATE 2014 among (approximately) 1,55,190 students in India.
11. Ministry of Human Resource and Development (MHRD) Scholarship for Postgraduate education, India.
12. Best Outgoing student in Computer Science and Engineering (P Rathnaswamy Memorial Endowment) for the year 2014.
13. First prize in Coding Competition, CODESTORM, conducted by IEEE Computer Society (March 2013).
14. Received Ashok Leyland "ALL THE BEST" Scholarship for graduate studies, India (February 2011).
15. Recipient of Federal Bank Hormis Memorial Foundation Scholarship for graduate studies, India (2010-11).
16. Recipient of Indian Oil Education Scholarship for graduate education, India (2010-11).
17. Received Central Sector Scholarship by Department of Higher Education (MHRD) for graduate studies, India (2010).
18. Achieved a rank of 629 in Kerala Engineering Entrance (KEAM 2010) among (approximately) 100949 students.
19. Received Malayala Manorama Merit Scholarship (February 2008).
20. Received prize at CMS Math Prodigy Hunt 2009, organized by Centre for Research in Mathematics.
21. Participated in 20th Kerala Science Congress, Trivandrum (January 2008).
22. Participated in Youth Parliament Competition under the auspices of the Institute of Parliamentary Affairs, Government of Kerala.

Skills

Programming	C/C++, Java, Javascript, Python, PyTorch
DevOps	AWS, Docker, Jenkins, Jira, SonarQube
DBMS	Oracle, SQL
Web	HTML5, CSS3, jQuery, JSP
Tools	NetBeans, Eclipse, VS Code, LaTeX

Personal Data

Born	24th April 1992 in Kerala, India
Citizenship	Indian
Marital Status	Married
Languages	Malayalam (mother tongue), English, Hindi, Tamil
Interests	Photography, Badminton, Cycling
Contact	Email: ajith424suresh@gmail.com, Mob.: (+91) 8762049224